

CHEMISTRY**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 8 printed pages.
4. It is divided into four sections and has twenty one questions in all.
5. Answer all questions.
6. Section A has fourteen subparts. Each question carries 1 mark.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B has ten questions. Each question carries 2 marks.
9. Section C has seven questions. Each question carries 3 marks.
10. Section D has three questions. Each question carries 5 marks.
11. Internal choices have been provided in two questions each in Sections B, C and D.
12. The intended marks for questions are given in brackets [].
13. All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.
14. Balanced equations must be given wherever possible and diagrams where they are helpful.
15. When solving numerical problems, all essential workings must be shown.
16. In working out problems, use the following data:
Gas constant $R = 1.987 \text{ cal deg}^{-1} \text{ mol}^{-1} = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$, $1 \text{ l atm} = 1 \text{ dm}^3 \text{ atm} = 101.3 \text{ J}$.
 $1 \text{ faraday} = 96500 \text{ coulombs}$, Avogadro's number $= 6.023 \times 10^{23}$.

Instruction to Supervising Examiner*Kindly read aloud the Instructions given above to all the candidates present in the examination hall.***SECTION A – 14 MARKS****Question 1**

- (A) Fill in the blanks by choosing the appropriate word(s) from those given in the brackets.

[zinc, sodium, Clemmensen, Wolff-Kishner, phosgene, ethanol, methanol, five, four, completely, partially, 2,4,6-tribromophenol, 4-bromophenol, 3-bromophenol, carbon tetrachloride, half]

- (i) Acetophenone is reduced to ethylbenzene by _____ amalgam and concentrated hydrochloric acid. The reaction is known as _____ reduction. (Recall)

[1]

- (ii) Chloroform is stored in closed dark bottles filled to the brim, because in air and light it is oxidised to the poisonous gas _____; a little _____ is added to it to convert this gas into harmless ethyl carbonate. (*Understand*) [1]
- (iii) Among the bivalent ions of the first transition series, Mn^{2+} ($Z = 25$) has the highest magnetic moment because it has _____ unpaired electrons, whereas Zn^{2+} ($Z = 30$) salts are white because its d subshell is _____ filled. (*Analyse*) [1]
- (iv) Phenol on treatment with bromine water gives a white precipitate of _____, whereas with bromine in CS_2 at low temperature it gives mainly _____. (*Recall*) [1]
- (B) Select and write the correct alternative from the choices given below.
- (i) Which of the following aqueous solutions has the lowest vapour pressure at the same temperature? (Assume complete dissociation of the electrolytes.) (*Evaluate*) [1]
- (a) 0.1 m glucose
(b) 0.1 m NaCl
(c) 0.2 m urea
(d) 0.1 m $\text{Al}_2(\text{SO}_4)_3$
- (ii) Read the following statements: [1]
- P: Sucrose is a non-reducing sugar.
Q: Maltose on hydrolysis gives two molecules of α -D-glucose.
R: DNA contains the base uracil.
S: The primary structure of a protein is destroyed on denaturation.
- Which of the statements are correct?
(*Evaluate*)
- (a) P and Q only
(b) P, Q and R only
(c) Q and S only
(d) P and S only
- (iii) Which of the following complexes is tetrahedral and paramagnetic? (Atomic number of Ni = 28, Pt = 78) (*Analyse*) [1]
- (a) $[\text{Ni}(\text{CO})_4]$
(b) $[\text{Ni}(\text{CN})_4]^{2-}$
(c) $[\text{NiCl}_4]^{2-}$
(d) $[\text{PtCl}_4]^{2-}$
- (iv) For the reaction $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$, the rate law is $\text{rate} = k[\text{NO}]^2[\text{O}_2]$. If the volume of the reaction vessel is suddenly reduced to half, the rate of the reaction becomes: (*Apply*) [1]
- (a) one half of the original rate
(b) two times the original rate
(c) four times the original rate
(d) eight times the original rate

- (v) Which of the following gives an offensive-smelling compound when warmed with chloroform and alcoholic KOH? (*Understand*) [1]
- (a) $(\text{CH}_3)_2\text{NH}$
 (b) $(\text{C}_2\text{H}_5)_3\text{N}$
 (c) $\text{C}_6\text{H}_5\text{NH}_2$
 (d) $\text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$
- (vi) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]
- Assertion:** Anisole on heating with HI gives phenol and methyl iodide.
Reason: The O–C(aryl) bond has partial double-bond character, so the iodide ion attacks the methyl carbon and not the ring carbon.
- (*Analyse*)
- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
 (c) Assertion is true and Reason is false.
 (d) Both Assertion and Reason are false.
- (vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]
- Assertion:** The limiting molar conductivity of acetic acid cannot be obtained by extrapolating its Λ_m vs $\sqrt{c}$ graph to zero concentration.
Reason: By Kohlrausch's law, the limiting molar conductivity of an electrolyte is the sum of the limiting ionic conductivities of its cation and anion.
- (*Analyse*)
- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
 (c) Assertion is true and Reason is false.
 (d) Both Assertion and Reason are false.
- (C) Read the passage carefully and answer the questions that follow.
- Osmosis is the flow of solvent through a semipermeable membrane from a dilute solution to a more concentrated one. The excess pressure that must be applied to the solution to stop this flow is its osmotic pressure, $\pi = CRT$. A 0.9 % (mass/volume) sodium chloride solution, called normal saline, has the same osmotic pressure as the fluid inside red blood cells, and is used for intravenous injection. A red blood cell placed in 5 % NaCl solution shrivels, while one placed in pure water swells and may burst.
- Because π can be measured accurately at room temperature even for very dilute solutions, it is the method of choice for finding the molar masses of proteins and polymers. If a pressure larger than the osmotic pressure is applied to the solution, pure solvent is pushed out through the membrane; this reverse osmosis is used to desalinate sea water.
- (i) Name the type of solution that 5 % NaCl is with respect to the cell fluid, and explain why the red blood cell shrivels in it. (*Understand*) [1]

- (ii) Give two reasons why osmotic pressure is preferred to other colligative properties for determining the molar mass of a protein. (*Evaluate*) [1]
- (iii) What is reverse osmosis? Name a material used as the membrane in desalination plants. (*Understand*) [1]

SECTION B – 20 MARKS

Question 2

The standard reduction potentials of some electrodes are given below:

[2]

Electrode	E° (V)
K^+/K	-2.93
Al^{3+}/Al	-1.66
Ni^{2+}/Ni	-0.25
Hg^{2+}/Hg	+0.85
Au^{3+}/Au	+1.50

(a) Arrange the cations in increasing order of their oxidising power. (b) Which of these metals will not liberate hydrogen from dilute hydrochloric acid? Justify.

(*Apply*)

Question 3

Give one chemical test each to distinguish between the following pairs of compounds. Write the chemical equation for the reaction involved.

- (i) Propanal and propanone (*Analyse*) [1]
- (ii) Benzaldehyde and acetaldehyde (*Analyse*) [1]

Question 4

- (i) Answer the following for the complex ions $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ (atomic number of Co = 27).

- (a) State the hybridisation of cobalt in each. (*Understand*) [1]
- (b) Which of the two is an outer-orbital complex? Comment on the magnetic nature of each. (*Analyse*) [1]

OR

- (ii) Two students were asked about the complex ion $[Ni(CN)_4]^{2-}$ (atomic number of Ni = 28). Riya said that it is tetrahedral and paramagnetic; Kabir said that it is square planar and diamagnetic.
- (a) Whose answer is correct? (*Evaluate*) [1]
- (b) Justify your answer using valence bond theory. (*Analyse*) [1]

Question 5

The cobalt-60 source used in a hospital for radiotherapy has a half-life of 5.3 years. The source is replaced when 90 % of it has decayed. After how many years must the source be replaced? (Radioactive decay follows first order kinetics.) (*Apply*) [2]

Question 6

How can the following conversions be carried out? (Write the chemical equations.)

- (i) Aniline to benzonitrile (*Recall*) [1]
- (ii) Propene to 1-bromopropane (*Recall*) [1]

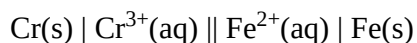
Question 7

Substantiate the following statements with reasons.

- (i) Mn^{2+} compounds are more stable than Fe^{2+} compounds towards oxidation to the +3 state. [1]
(Atomic number of Mn = 25, Fe = 26) (*Understand*)
- (ii) Cu^{2+} compounds are paramagnetic whereas Cu^+ compounds are diamagnetic. (Atomic number of Cu = 29) (*Understand*) [1]

Question 8

Calculate the standard Gibbs energy change for the cell reaction of the cell given below, and state whether the reaction is spontaneous. [2]



Given: $E^\circ(\text{Cr}^{3+}/\text{Cr}) = -0.74 \text{ V}$, $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$ (*Apply*)

Question 9

- (i) Illustrate the following reactions with an example in each case:
 - (a) Carbylamine reaction (*Recall*) [1]
 - (b) Coupling reaction of benzenediazonium chloride with aniline (*Recall*) [1]

OR

- (ii) Three isomeric amines [P], [Q] and [R] have the molecular formula $\text{C}_3\text{H}_9\text{N}$. [P] reacts with benzenesulphonyl chloride to give a product soluble in aqueous KOH; [Q] gives a product insoluble in KOH; [R] does not react. [P] has an unbranched chain. Identify [P], [Q] and [R], and write the reaction of [P]. (*Analyse*) [2]

Question 10

The rate of a reaction doubles when the temperature is raised from 300 K to 310 K. Calculate the energy of activation of the reaction. (*Apply*) [2]

Question 11

Give one chemical test to distinguish between the following pairs of compounds:

- (i) Propan-1-ol and propan-2-ol (*Analyse*) [1]
- (ii) Propan-2-ol and 2-methylpropan-2-ol (*Analyse*) [1]

SECTION C – 21 MARKS

Question 12

Write the chemical equation and name the reaction in each of the following:

- (i) Methyl bromide is heated with silver fluoride. (*Recall*) [1]
- (ii) Ethyl chloride is treated with sodium iodide in dry acetone. (*Recall*) [1]

- (iii) Chlorobenzene is treated with chlorine in the presence of anhydrous ferric chloride. [1]
(Understand)

Question 13

For the reaction $2\text{NO(g)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{NOCl(g)}$, the following data were collected at a constant temperature:

Experiment	[NO] (mol L ⁻¹)	[Cl ₂] (mol L ⁻¹)	Initial rate (mol L ⁻¹ s ⁻¹)
1	0.010	0.010	1.2×10^{-4}
2	0.010	0.020	2.4×10^{-4}
3	0.020	0.020	9.6×10^{-4}

- (i) Find the order with respect to each reactant and write the rate law. (Analyse) [1]
 (ii) Calculate the rate constant, with its unit. (Apply) [1]
 (iii) Calculate the initial rate when [NO] = 0.030 mol L⁻¹ and [Cl₂] = 0.040 mol L⁻¹. (Apply) [1]

Question 14

- (i) Answer the following:
 (a) Arrange in increasing order of acid strength: water, ethanol, phenol, 4-nitrophenol. (Analyse) [1]
 (b) Why is 4-nitrophenol a stronger acid than phenol? (Evaluate) [1]
 (ii) Ethyl *tert*-butyl ether is prepared from sodium *tert*-butoxide and ethyl bromide, and not from sodium ethoxide and *tert*-butyl bromide. Explain. (Evaluate) [1]

OR

- (iii) How will the following be converted? (Give chemical equations.)
 (a) Phenol to salicylic acid (Recall) [1]
 (b) Propene to propan-1-ol (Recall) [1]
 (c) Chlorobenzene to phenol (Recall) [1]

Question 15

- (i) Complete and balance: $\text{KMnO}_4 + \text{KI} + \text{H}_2\text{SO}_4 \rightarrow \text{_____} + \text{_____} + \text{_____} + \text{_____}$ (Understand) [1]
 (ii) Complete and balance: $\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 + \text{H}_2\text{S} \rightarrow \text{_____} + \text{_____} + \text{_____} + \text{_____}$ (Understand) [1]
 (iii) State two consequences of lanthanoid contraction. (Analyse) [1]

Question 16

- (i) Why is the product of hydrolysis of sucrose called invert sugar? (Understand) [1]
 (ii) Distinguish between fibrous and globular proteins, giving one example of each. (Understand) [1]
 (iii) Name the vitamin whose deficiency causes (a) scurvy (b) pernicious anaemia. (Recall) [1]

Question 17

- (i) The limiting molar conductivities of HCl, NaCl and CH₃COONa are 426.2, 126.4 and 91.0 S cm² mol⁻¹ respectively at 298 K.
 (a) Calculate the limiting molar conductivity of acetic acid. (Apply) [1]

- (b) The conductivity of 0.01 M acetic acid is $1.63 \times 10^{-4} \text{ S cm}^{-1}$. Calculate its molar conductivity and degree of dissociation. (*Apply*) [2]

OR

- (ii) The resistance of a conductivity cell filled with 0.1 M KCl solution is 100 Ω . When the same cell is filled with a 0.02 M solution of an electrolyte, its resistance is 520 Ω . Calculate (a) the cell constant, (b) the conductivity and (c) the molar conductivity of the 0.02 M solution. (Conductivity of 0.1 M KCl = 0.0129 S cm^{-1}) (*Apply*) [3]

Question 18

Write the chemical equations for the following conversions: (*Recall*)

- (i) Aniline to iodobenzene [1]
 (ii) Ethanamine to ethanol [1]
 (iii) Benzoic acid to aniline [1]

SECTION D – 15 MARKS

Question 19

- (i) An organic compound [A], $\text{C}_3\text{H}_6\text{O}$, does not reduce Tollens' reagent but gives a yellow precipitate with iodine and sodium hydroxide. [A] reacts with methyl magnesium bromide followed by hydrolysis to give [B]. [A] on reduction with zinc amalgam and concentrated HCl gives [C]. Two molecules of [A] condense in the presence of barium hydroxide to give [D], $\text{C}_6\text{H}_{12}\text{O}_2$.
 (a) Identify [A], [B], [C] and [D]. (*Understand*) [2]
 (b) Write the equation for the formation of [D] and name the reaction. (*Recall*) [1]
 (c) Give a chemical test to distinguish [A] from propanal. (*Analyse*) [1]
 (d) Which is more reactive towards nucleophilic addition, [A] or propanal? Why? (*Evaluate*) [1]

OR

- (ii) Answer the following:
 (a) Identify [A], [B], [C] and [D]: [2]

$$\text{C}_6\text{H}_5\text{COOH} \xrightarrow{\text{SOCl}_2} [\text{A}] \xrightarrow{\text{H}_2, \text{Pd-BaSO}_4} [\text{B}] \xrightarrow{\text{conc. NaOH}} [\text{C}] + [\text{D}]$$

 (*Analyse*)
 (b) Arrange in increasing order of acid strength, giving a reason: CH_3COOH , ClCH_2COOH , FCH_2COOH , Cl_2CHCOOH . (*Evaluate*) [1]
 (c) Write chemical equations to convert: (1) acetic acid to acetone (2) benzene to acetophenone. (*Recall*) [2]

Question 20

- (i) For the complex ion $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (en = ethane-1,2-diamine):
 (a) Write its IUPAC name. (*Recall*) [1]
 (b) Draw its geometrical isomers and state which of them is optically active. (*Analyse*) [1]
 (ii) What type of isomerism is shown by $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$? Give a chemical test to distinguish them. (*Analyse*) [1]

- (iii) One mole of $\text{CoCl}_3 \cdot 5\text{NH}_3$ gives two moles of AgCl with excess of silver nitrate. Write the structural formula of the complex and name it. (*Evaluate*) [1]
- (iv) How is EDTA used in the treatment of lead poisoning? (*Understand*) [1]

Question 21

- (i) Answer the following:
- (a) 200 cm^3 of an aqueous solution of a protein contains 1.26 g of the protein. The osmotic pressure of the solution at 300 K is $2.57 \times 10^{-3} \text{ atm}$. Calculate the molar mass of the protein. (*Apply*) [2]
- (b) Acetic acid dimerises in benzene. A solution of 0.2 g of acetic acid in 20 g of benzene lowers its freezing point by 0.45 K. Calculate the degree of association of acetic acid. (K_f for benzene = $5.12 \text{ K kg mol}^{-1}$, molar mass of $\text{CH}_3\text{COOH} = 60 \text{ g mol}^{-1}$) (*Apply*) [2]
- (c) The molar mass of NaCl determined by the elevation of boiling point method is found to be nearly half its formula mass. Explain. (*Evaluate*) [1]

OR

- (ii) Answer the following:
- (a) A solution is made by dissolving 31 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) in 90 g of water. Calculate the mole fraction of each component and the molality of the solution. ($\text{C} = 12, \text{H} = 1, \text{O} = 16$) (*Apply*) [2]
- (b) A 5 % (mass/volume) solution of glucose is isotonic with a 1.66 % (mass/volume) solution of a non-volatile, non-electrolyte substance X at the same temperature. Find the molar mass of X. (*Apply*) [2]
- (c) What is an azeotrope? What type of azeotrope is formed by a solution showing negative deviation from Raoult's law? (*Understand*) [1]

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